James Han

310-755-9222 — jjhan@ucsd.edu — linkedin.com/in/jameshan310 — github.com/hamesjan

Summary — Visionary passionate about ideation, business management, mindful design/development, and execution.

Experience

Geltech - Lead Firmware/Software Engineer — **October 2024 – Present**
C, STM32, Python, RS232/485, SPI, I2C, USART, Autodesk Fusion
- Sole firmware/software developer for the control system of the Carousel hydrogel testing product, overseeing pre-alpha to pilot production.
- Design and implement core state machine that integrates fluidics, heating, mechanical, weighing, and control systems into a cohesive product.
- Perform critical component selection, system design, de-risking strategies and create high level system documents, block diagrams, and specification sheets for manufacturing partners.

Hyundai Mobis OTA Embedded/Software Engineer Intern — **Jul 2024 – Aug 2024**
Rust, C++, Java, Linux Kernel, Bash, Ubuntu, Markdown, Doxygen, Android (AOSP)
- Research In-place, Block-based, A/B dynamic partitions, EXT4 filesystem, directory, multi-image, differential patch file generation and full/regular update testing and development
- System profiling and trace analysis for IVI USB Update pipeline with atrace; bottom-up userspace tracing for optimization within updating process.
- Improved USB update process time by 54% by optimizing the chunk size (1MB to 32MB) sent in extracting/copying update file loop to reduce context switching.

Projects

MIPS Processor Optimization — **Jan 2025 - Mar 2025**
System Verilog
- Implemented four optimizations to the classical Hennesey and Patterson 5-Stage Processor.
- Implemented instruction cache next-line prefetching, data cache stride prefetching with stream buffers, branch prediction with global pattern tracking and fetch-stage branch target buffer, and two-way superscalar processing.
- Tracked and measured CPI, branch prediction accuracy, prefetching hits, data/instruction cache misses, and parallel instruction issue against nqueens, coin, esift2, and quickSort benchmarks to see improvements over baseline.

RaytechX — **June 2024**
Python
- Developed antenna design analysis automation system for collecting S12 data with a CNC Machine (Shapeoko XXL) and a VNA (Anritsu Ms4647B)
- Implemented automation with Raspberry Pi 5
- Executed sequential serial and TCP communication with two machines to coordinate CNC movement, Antenna positioning on two axes, and VNA data collection

Tri-tone — **Mar 2024 - June 2024**
C++, Python
- Built an embedded system with ESP32-S3, Teensy3.6, Audioshield, and Adafruit 64x64 RGB Matrix
- Used Pytorch to pre-train 2 hidden layer Feed-Forward Neural Network, automated data collection by generation of RGB labels for spectrogram data using ChatGPT
- Processes audio into 1024 bin FFT data and then sends to a pre-trained ML model for RGB inferencing and real-time color prediction based on current dominant frequency and musical context (10 previous dominant frequencies)
- Spectrogram visualization with separate mic
- Used OnShape CAD software to build enclosure, powered by 4A, 5v power supply with adapter and switch to power ESP, soldered ESP32 together with solder paste and stencil and soldered some connections by hand

Plots Inc. - Founder — **Aug 2021 - Feb 2022**
Flutter, Dart, Node.js, Firebase, GCP
- Independently designed, developed, published, and sold a party organization and discovery app using Flutter and Firebase, initially worked with current CEO/COO before acquisition
- App features: guest monitoring, dynamic ticketing system, host-guest messaging, security management, locational notification system and announcements.
- Software acquired in February 24, 2022, currently has over 100,000 users

Education

University of California - San Diego — **Aug 2021- Expected Jun 2026**
BS/MS in Computer Science and Engineering